

Emerging Role of Tele-Rehabilitation in Sports Physiotherapy during COVID 19 Pandemic

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World health organization (WHO) declared the COVID 19 a health emergency at international level in January 2020 and declared pandemic in March 2020 as it spread in almost every country of the world.¹ COVID 19 has become a public health crisis that has spread to most regions/countries of the world. Various preventive measures like social distancing, regular hand washing, use of alcohol based hand sanitizer, mandatory use of face mask, social distancing, avoidance of going to crowded/public places, etc. were followed worldwide.²

In the midst of this pandemic, rehabilitation became outrageously significant and assumed essential role in health care sector. Rehabilitation became crucial not only for patients with positive COVID 19 virus but for those also who were not infected with the virus.³ Proper rehabilitation helps to promote better health and functioning, leading towards faster recovery and prompt discharge from hospitals and prevent relapse and deterioration of condition after discharge. For patients with explicit conditions without COVID-19, recovery has been perceived as a fundamental need, whose deferral or stoppage can prompt a

decrease in health related quality of life outcomes.⁴

A number of international organizations/stakeholders including the World Confederation of Physical Therapists (WCPT), European Respiratory Society (ERS), American Thoracic Society (ATS), and Chartered Society of Physiotherapy (CSP) have published guidelines and position papers on the role of physiotherapy and rehabilitation in managing COVID-19.⁵ Considering all these guidelines physiotherapy plays a frontline role in the acute and even post infective rehabilitation phases of COVID-19 patients. As also it is necessary to maintain the continuation of rehabilitation for people with disabilities and the elderly with appropriate modifications to the delivery of services to ensure safety as the pandemic demands.⁶

During pandemic physiotherapists adopted various methods to limit spread of infection like limited face to face consultation, strict adherence to infection prevention and control guidelines, use of personal protective equipments (PPE) and psychological support strategies for frontline healthcare professionals to ensure the mental health

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during pandemic. One of the important innovations in the rehabilitation field is the use of remote consultation which is also known as telehealth, telemedicine or telerehabilitation that ensures continuous and safe delivery of services.⁷

Tele-rehabilitation refers specifically to clinical rehabilitation services with the focus of evaluation, diagnosis, and treatment. Tele-rehabilitation has been employed as a catch-all term for digital rehabilitation methods and is variously called digital health, eHealth, or mobile health.⁸ Telerehabilitation can be provided in a variety of ways, including 2-way real-time visits with audio, video, or both; asynchronous e-visits; virtual check-ins; remote evaluations of recorded videos or images; and telephone assessment and management services.⁸ Telehealth encompasses preventive, promotive and curative aspects of health and many different healthcare professions are involved in its delivery. The key benefit of telerehabilitation is that it enables personalized consultations and treatments to continue via electronic media (phone and/or video) during the pandemic. This reduces the risk of transmission of the virus. This is significant as it protects both patients and the healthcare team.⁹

Sports training and rehabilitation had a major setback with majority of sports training arenas, stadia and gyms being shut to avoid the aggregation of

people fearing the spread of COVID-19 pandemic. In addition, there were no competitive sporting events, hence the training calendar of the athletes got disrupted. It is obvious that not only physical, but also mental health of the athletes suffered without structured routine. Especially in terms of an injured athlete post acute trauma or surgery or even with chronic musculo-skeletal issues, there may be loss of physical function with residual disability and psychological impact of being out of action. Also return to sports with elimination of risk of re-injury becomes a bigger challenge due to lack of monitoring of the injury and recovery status. Tele-rehabilitation interventions could provide an option to address this rehabilitation gap.¹⁰ It can provide evidence-based information, education, and personalized exercise guidance with regular monitoring and goal setting.¹¹ Evaluation and management of injured athletes is feasible through tele-rehabilitation as well.¹²

Not only with reference to the present global health crisis, but there has been significant amount of ongoing research reviewing the efficacy of tele-rehabilitation for musculo-skeletal physiotherapy and sports rehabilitation in the recent past. A current review stated that pain relief, functionality measures and health related quality of life were better with tele-rehabilitation than in person rehabilitation in musculo-skeletal physical therapy.¹³ In a reliability study,

where online assessment of knee complex was carried out where the subjects had to perform facilitated self-palpation, self-applied modified orthopaedic tests, active movements and functional tasks. It was found that online assessment was in agreement with 87% of cases for the patho-anatomical diagnosis identified by physical examination.¹⁴ Similarly in remote assessment of subjects with pain in ankle joint complex, it was found that tele-rehabilitation showed 93.3% agreement in patho anatomical diagnosis and 80% agreement with the primary system of diagnosis in comparison with face to face rehabilitation.¹⁵ In assessment and diagnostic tests of elbow joint complex, tele-rehabilitation showed 73% validity and 90% reliability in comparison to conventional face to face physical examination tests.¹⁶ In a 3 arm parallel group randomized control trial studying rotator cuff related shoulder pain (RCRSP), where subjects were allotted to 3 groups- advice only (rotator cuff and shoulder related education and activity advice), advice and with recommended care (information about exercises and other treatment for RCRSP) and advice and with recommended care and telerehabilitation (along with exercise advice there was weekly teleconferencing physiotherapy session for guided self-exercise plan and communication with subjects).¹⁷ The group with tele-rehabilitation showed around 70% adherence to weekly

exercise program, better responsiveness to periodic online assessments and better pain and shoulder function outcomes as compared to the other 2 groups without tele-rehabilitation.¹⁷ Tele-rehabilitation was utilized for rehabilitation of adults with less than 5 years history of ACL reconstruction wherein 70% were competitive or recreational athletes. It was concluded that the participation and completion rates were almost 96% and 77% felt satisfied with their progress after an average of 6 months of tele-rehabilitation.¹⁸ 55% of the subjects reported return to sport with the pre injury level of knee function; however the risk of re-injury rates would be high as they had not satisfied the return to sports criteria on direct monitoring.¹⁸

Apart from the reviewed evidence, expert opinions of the sports science staff and coaches utilizing tele-rehabilitation also have fairly positive responses. The developmental or non elite young athletes were effectively kept engaged with online educational and practical sessions by the coaches and sports science experts even if they were far from the training arenas without direct personal contact. It also gave the athletes an opportunity to focus on self-reflection and improvement of individual issues and address them personally with the concerned experts which otherwise would be difficult during hectic on-field training. On the other hand, there were certain pitfalls noticed like non-compliance of certain

athletes which were low on motivation or had certain lack of economic stability and digital communication awareness, and could not be tracked easily also. Lack of periodization and time frame maintenance in injury rehabilitation or specific conditioning was another issue if the athlete failed to keep up with the exercise intricacies like technique or volume or progression. The remoteness of the location of certain athletes and lack of digital literacy was also a hindrance. In case of elite athletes preparing for upcoming international events, they were not mostly considered eligible for tele-rehabilitation. They had to continue their on-field training with lot of unfavourable circumstances like limited training hours, lack of sufficient support staff, less opponents to practice with, restricted travel, non-accessibility to gyms etc. This might have definitely affected their morale, confidence, physical and mental preparedness to a certain extent when they participate in a competitive sporting event.

There are various modalities that can be used for delivering tele-rehabilitation:-¹⁹

- Virtual reality- Mimicking the real world environment with interactive portals for education, training, rehabilitation.
- Video-conferencing- Via online communication platforms like Zoom, Google Meet, and Microsoft teams for face to face interactions between consultants themselves or consultants and people. Either can be one to one or

group basis.

- Interactive Mobile Apps- Provides a Questionnaire for self-assessment of an individual condition and also guides the people for home management or necessary consultation
- Exercise and Fitness portals- Delivers exercise videos and educational guidance as per the fitness requirement or the nature of physical problem of the individual.
- Webinar and E-education series- Continuous education program for knowledge enhancement and engagement of a common interest group (Eg- athletes).

Benefits of tele-rehabilitation include:

- Innovative technology is often appealing to users, which can increase adherence and decrease the likelihood that people will stop treatment.
- Direct contact between people with physiotherapist and the ability for constant feedback enhances trust, as well as peace of mind.
- Expanded peoples' reach or access particularly for those living in provincial regions or for those who have limited access to health care services.
- Reduced rates of missed visits and delayed consultation issues.
- Travel related time and expenses can be reduced.
- People can access the services in various areas like homes, offices etc.
- Personal satisfaction with tele-rehabilitation is equivocal to traditional care.
- Reduces clinical burden on the

physiotherapist in terms of time required for treatment and ease of accessing multiple patients simultaneously.

- Most importantly minimizes unnecessary physical contact between physiotherapist and patient effectively reducing risk of infection transmission (Eg- need for social distancing in COVID 19)

However there are various challenges in the field of tele-rehabilitation which is still being faced and need to be addressed.

- Tele-rehabilitation eliminates the physical contact which usually exists between the physiotherapist and the patient during the sessions of treatment. Some of the popular purposes of touch include physical examination (such as in palpation), provision of manual intervention (such as in myofascial release, mobilizations), and assistive touch.²⁰

- There is no sufficient amount of evidence to support the outcome and effectiveness of tele-rehabilitation and in person physiotherapy sessions.

- Internet connectivity challenges and digital literacy are another major problem in rural areas and developing countries.²¹

Various research investigations have

upheld and prescribed the utilization of tele-rehabilitation for physiotherapy service delivery. With the current worldwide pandemic, numerous physiotherapy administrations across various nations have to depend on it. Despite some limitations, tele-rehabilitation offers a protected and viable option in contrast to conventional in-person care, particularly during time of emergency as seen in COVID-19, when safety measures should be taken to forestall transmission of disease. Tele-rehabilitation remains widely suggested for the propagation of sports rehabilitation especially for athletes who require continuous education, motivation, progression, monitoring physical and mental conditioning under the circumstances. However, from the existing evidence and practical experience we recommend that tele-rehabilitation can be used with discretion in sports interspersed with regular in person contractual sessions as per the feasibility and safety of the circumstance. This may ensure better adherence of the athlete and may also address the necessity for certain testing and interventions that require personal intervention.

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